

$$\begin{aligned}
& \frac{1}{m_s^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} + \frac{1}{16\pi^2} \left(\frac{1}{24} \frac{1}{m_b^2} s_\gamma^2 \left(\overline{y}_d^{i4i3} \left(9 \overline{y}_e^{pr} y_e^{pi2} y_e^{i1r} (1 + s_\gamma^2) + 2 y_e^{i1i2} (11 g_1^2 + \right. \right. \right. \\
& \quad \left. \left. \left. 9 g_2^2 + 24 g_3^2 - 6 c_\gamma^2 (3 \overline{y}_d^{pr} y_d^{pr} + \overline{y}_e^{pr} y_e^{pr}) \right) \right) + \right. \\
& \quad \left. \left. 3 y_e^{i1i2} (3 \overline{y}_d^{ri3} \overline{y}_d^{i4p} y_d^{rp} (1 + s_\gamma^2) - \overline{y}_d^{pi3} \overline{y}_u^{i4r} y_u^{pr} (-1 + c_\gamma^2) \right) \right) + \\
& \frac{1}{2} \sum_p s_\gamma g_1^2 \frac{1}{m_s^4} \overline{y}_d^{i4i3} y_e^{i1i2} (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) \text{LF}_{1,0}[m_d^P] + \\
& 3 s_\gamma \frac{1}{m_s^4} \overline{y}_d^{pr} \overline{y}_d^{i4i3} y_d^{pr} y_e^{i1i2} (-s_{2\gamma} c_\gamma + s_\gamma^3) \text{LF}_{1,0}[m_d^r] + \\
& \frac{1}{2} \sum_p s_\gamma g_1^2 \frac{1}{m_s^4} \overline{y}_d^{i4i3} y_e^{i1i2} (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) \text{LF}_{1,0}[m_e^P] + \\
& s_\gamma \frac{1}{m_s^4} \overline{y}_d^{i4i3} \overline{y}_e^{pr} y_e^{pr} y_e^{i1i2} (-s_{2\gamma} c_\gamma + s_\gamma^3) \text{LF}_{1,0}[m_e^r] + \\
& \frac{1}{2} s_\gamma \frac{1}{m_s^4} \overline{y}_d^{i4i3} y_e^{i1i2} (2 \overline{y}_e^{pr} y_e^{pr} (-s_{2\gamma} c_\gamma + s_\gamma^3) - \sum_p g_1^2 (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma})) \text{LF}_{1,0}[m_l^P] + \\
& \frac{1}{2} s_\gamma \frac{1}{m_s^4} \overline{y}_d^{i4i3} y_e^{i1i2} \\
& \quad (6 \overline{y}_d^{pr} y_d^{pr} (-s_{2\gamma} c_\gamma + s_\gamma^3) + 6 c_\gamma \overline{y}_u^{pr} y_u^{pr} (s_{2\gamma} + s_\gamma c_\gamma) + \sum_p g_1^2 (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma})) \\
& \quad \text{LF}_{1,0}[m_q^P] - \sum_p s_\gamma g_1^2 \frac{1}{m_s^4} \overline{y}_d^{i4i3} y_e^{i1i2} (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) \text{LF}_{1,0}[m_u^P] + \\
& 3 s_\gamma c_\gamma \frac{1}{m_s^4} \overline{y}_d^{i4i3} y_e^{i1i2} \overline{y}_u^{pr} y_u^{pr} (s_{2\gamma} + s_\gamma c_\gamma) \text{LF}_{1,0}[m_u^r] + \frac{1}{4} s_\gamma \frac{1}{m_s^4} \overline{y}_d^{i4i3} y_e^{i1i2} \\
& \quad (3 s_{4\gamma} c_\gamma (g_1^2 + g_2^2) + s_\gamma (g_1^2 (-1 + 3 c_{2\gamma}^2) + 3 g_2^2 (-1 + c_{2\gamma}^2))) \text{LF}_{1,0}[m_\Phi] + \\
& \frac{1}{4} \frac{1}{m_s^2} s_\gamma^2 \left(\overline{y}_d^{i4i3} (-3 s_\gamma^2 \overline{y}_e^{pr} y_e^{pi2} y_e^{i1r} + 2 y_e^{i1i2} (g_1^2 + 3 g_2^2)) + \right. \\
& \quad \left. 3 y_e^{i1i2} (-s_\gamma^2 \overline{y}_d^{ri3} \overline{y}_d^{i4p} y_d^{rp} + c_\gamma^2 \overline{y}_d^{pi3} \overline{y}_u^{i4r} y_u^{pr}) \right) \text{LF}_{1,1}[m_\Phi] - \\
& \frac{1}{4} s_\gamma^2 y_e^{i1i2} (\overline{y}_d^{i4i3} (g_1^2 + 3 g_2^2) + 2 c_\gamma^2 \overline{y}_d^{pi3} \overline{y}_u^{i4r} y_u^{pr}) \text{LF}_{1,2}[m_\Phi] - \\
& \frac{1}{9} g_1^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,0}[m_1, m_d^{i3}] + \frac{1}{18} g_1^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,1,-1}[m_1, m_d^{i3}] - \\
& g_1^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,0}[m_1, m_e^{i2}] + \frac{1}{2} g_1^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,1,-1}[m_1, m_e^{i2}] - \\
& \frac{1}{4} g_1^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,0}[m_1, m_l^{i1}] + \frac{1}{8} g_1^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,1,-1}[m_1, m_l^{i1}] - \\
& \frac{1}{36} g_1^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,0}[m_1, m_q^{i4}] + \\
& \frac{1}{72} g_1^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,1,-1}[m_1, m_q^{i4}] - g_1^2 \frac{1}{m_b^4} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,-1}[m_1, \tilde{\mu}] - \\
& 2 m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 \frac{1}{m_b^4} \overline{y}_d^{i4i3} y_e^{i1i2} (c_\gamma^2 - 2 s_\gamma^2) \text{LF}_{1,1,0}[m_1, \tilde{\mu}] - \\
& \frac{3}{4} g_2^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,0}[m_2, m_l^{i1}] + \\
& \frac{3}{8} g_2^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,1,-1}[m_2, m_l^{i1}] - \frac{3}{4} g_2^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,0}[m_2, m_q^{i4}] + \\
& \frac{3}{8} g_2^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,1,-1}[m_2, m_q^{i4}] - 3 g_2^2 \frac{1}{m_b^4} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,-1}[m_2, \tilde{\mu}] - \\
& 6 m_2 s_\gamma \tilde{\mu} c_\gamma g_2^2 \frac{1}{m_b^4} \overline{y}_d^{i4i3} y_e^{i1i2} (c_\gamma^2 - 2 s_\gamma^2) \text{LF}_{1,1,0}[m_2, \tilde{\mu}] - \\
& \frac{4}{3} g_3^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,0}[m_3, m_d^{i3}] + \frac{2}{3} g_3^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,1,-1}[m_3, m_d^{i3}] - \\
& \frac{4}{3} g_3^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,0}[m_3, m_q^{i4}] + \frac{2}{3} g_3^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,1,-1}[m_3, m_q^{i4}] - \\
& \frac{1}{2} \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{ri3} \overline{y}_d^{i4p} y_d^{rp} y_e^{i1i2} \text{LF}_{1,1,0}[m_d^P, \tilde{\mu}] + 3 s_\gamma \frac{1}{m_b^4} \overline{y}_d^{i4i3} y_e^{i1i2} \\
& \quad (s_\gamma \overline{a}_d^{pr} a_d^{pr} (-2 c_\gamma^2 + s_\gamma^2) - \tilde{\mu} c_\gamma (c_\gamma^2 - 2 s_\gamma^2) (\overline{y}_d^{pr} a_d^{pr} + y_d^{pr} \overline{a}_d^{pr}) + 3 s_\gamma \tilde{\mu}^2 c_\gamma^2 \overline{y}_d^{pr} y_d^{pr}) \\
& \quad \text{LF}_{1,1,0}[m_d^r, m_q^P] + s_\gamma \frac{1}{m_b^4} \overline{y}_d^{i4i3} y_e^{i1i2} \\
& \quad (s_\gamma \overline{a}_e^{pr} a_e^{pr} (-2 c_\gamma^2 + s_\gamma^2) - \tilde{\mu} c_\gamma (c_\gamma^2 - 2 s_\gamma^2) (\overline{y}_e^{pr} a_e^{pr} + y_e^{pr} \overline{a}_e^{pr}) + 3 s_\gamma \tilde{\mu}^2 c_\gamma^2 \overline{y}_e^{pr} y_e^{pr}) \\
& \quad \text{LF}_{1,1,0}[m_e^r, m_l^P] - \frac{1}{2} \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} \overline{y}_e^{pr} y_e^{pi2} y_e^{i1r} \text{LF}_{1,1,0}[m_e^r, \tilde{\mu}] + s_\gamma c_\gamma \frac{1}{m_b^2} \overline{y}_d^{i4i3} \\
& \quad y_e^{i1i2} (\tilde{\mu} \overline{y}_e^{pr} (a_e^{pr} (c_\gamma^2 - s_\gamma^2) - 2 s_\gamma \tilde{\mu} c_\gamma y_e^{pr}) + \overline{a}_e^{pr} (2 s_\gamma c_\gamma a_e^{pr} + \tilde{\mu} y_e^{pr} (c_\gamma^2 - s_\gamma^2))) \\
& \quad \text{LF}_{2,1,0}[m_l^P, m_e^r] + s_\gamma c_\gamma \frac{1}{m_b^2} \overline{y}_d^{i4i3} y_e^{i1i2} (\tilde{\mu} \overline{y}_e^{pr} (a_e^{pr} (-c_\gamma^2 + s_\gamma^2) + 2 s_\gamma \tilde{\mu} c_\gamma y_e^{pr}) + \\
& \quad \overline{a}_e^{pr} (-2 s_\gamma c_\gamma a_e^{pr} + \tilde{\mu} y_e^{pr} (-c_\gamma^2 + s_\gamma^2))) \text{LF}_{3,1,-1}[m_l^P, m_e^r] + \\
& c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} (c_\gamma \overline{a}_e^{pr} - s_\gamma \tilde{\mu} \overline{y}_e^{pr}) (c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr}) \text{LF}_{3,1,0}[m_l^P, m_e^r] - \\
& 3 c_\gamma^2$$